

Medicine Scan Web App – Complete Discussion & Plan

1. Idea Overview

The goal is to build a **med-tech web application** that helps users identify medicines and check their availability in nearby medical shops.

The app has **two main users**: - **Consumer/User** – scans medicine and views details + availability - **Shopkeeper** – updates medicine stock and availability

This project will be built first as a **small working prototype (MVP)** and expanded later.

2. Core Features (Finalized)

Consumer (User Web App)

- Scan medicine **using barcode**
- Scan medicine **without barcode** (photo/text scan)
- View medicine details:
 - Name
 - Uses / purpose
 - Diseases treated
 - Basic side effects
- View availability:
 - Which medical shops have stock
 - Quantity available
 - Availability status
 - Last updated time

 Informational purpose only. No dosage or prescription advice.

Shopkeeper Web App

- Login / Signup
 - Add medicine to shop inventory
 - Update stock quantity
 - Mark availability (in stock / out of stock)
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3. Scanning Methods Explained

1. Barcode Scanning (Primary & Mandatory)

- Uses camera to scan barcode
- Very accurate and fast
- Best for MVP
- Technologies:
 - QuaggaJS
 - ZXing JS

2. Scan Without Barcode (Advanced but Planned)

Option A: Text Scan using OCR (Recommended)

- User clicks photo of medicine strip/bottle
- OCR extracts text (medicine name, strength)
- System matches text with database
- User confirms medicine

Tools: - Tesseract.js (free) - Google Vision API (paid, later)

Option B: Image Recognition (Future Scope)

- AI recognizes medicine by appearance
- Requires large dataset & ML training
- Not included in prototype

Final Approach (Hybrid)

1. Try barcode scan
2. If barcode not found → scan image text (OCR)
3. User confirms result

4. Technology Stack (Chosen)

Frontend

- HTML
- CSS
- JavaScript

Backend

- Node.js
- Express.js

Database

- MongoDB (MongoDB Atlas – cloud)

Scanning

- Barcode: QuaggaJS
 - OCR: Tesseract.js
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5. Application Flow

User Flow

1. Open web app
2. Click **Scan Medicine**
3. Scan barcode OR capture image
4. Medicine identified
5. View medicine details
6. View shop availability and stock

Shopkeeper Flow

1. Login
 2. Open dashboard
 3. Add / update medicine
 4. Update quantity
 5. Save stock
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6. Database Structure (Simplified)

Medicine Collection

- medicine_id
- name
- barcode
- uses
- side_effects

Shop Collection

- shop_id
- shop_name
- location
- medicines:
- medicine_id

- quantity
 - last_updated
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7. Backend APIs (Planned)

- GET /medicine/barcode/:code → Get medicine by barcode
 - GET /medicine/search/:name → Get medicine by name
 - POST /shop/login → Shopkeeper login
 - POST /shop/add-medicine → Add/update stock
 - GET /medicine/availability/:id → Get shops with stock
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8. Step-by-Step Development Plan

Phase 1 – Planning

- Finalize features
- Draw app flow
- Decide tech stack

Phase 2 – Setup

- Create project structure
- Setup Node.js server
- Connect MongoDB

Phase 3 – Backend

- Create database models
- Build APIs
- Test APIs using Postman

Phase 4 – Shopkeeper App

- Login page
- Dashboard
- Add/update medicine stock

Phase 5 – Consumer App

- Home page
- Scan page
- Result page
- Availability page

Phase 6 – Scanning Features

- Barcode scanning integration
- OCR text scanning integration

Phase 7 – Testing & Safety

- Error handling
- Test edge cases
- Add medical disclaimer

Phase 8 – Deployment

- Frontend: Netlify / Vercel
 - Backend: Render
 - Database: MongoDB Atlas
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9. Legal & Safety Notes

- App is informational only
 - No diagnosis or dosage
 - User must consult doctor
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10. Future Enhancements

- Mobile app version
 - AI-based image recognition
 - Prescription scanning
 - Doctor & pharmacy integration
 - Online ordering
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11. Current Status

- Idea validated
 - Flow finalized
 - Ready to start technical implementation
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12. Next Steps (User Choice)

- Start database design
- Start backend coding
- Start barcode scanning

- Start OCR scanning
- Start UI development

This document captures the complete discussion and finalized plan for the Medicine Scan Web App prototype.